THEJAS DIXIT SATHYANARAYANA

Engineer – Robotics & Automation | Simulation & Application | Systems and Mechanical Design

Straubing, Germany | +49 15781288410

thejasds21@gmail.com | linkedin.com/in/thejas-dixit-s | github.com/Thejas12Dixit

PROFILE

Robotics and automation engineer with a research and applied engineering background in robot simulation, offline programming, virtual commissioning and mechanical design. M.Sc. in Systems Engineering for Manufacturing (OvGU Magdeburg). Research experience in PPR-based digital factory modeling, bin-picking simulation, Industry 4.0 data standards (AutomationML, AAS, OPC-UA), and multi-brand robot OLP across KUKA, ABB, FANUC, and UR platforms. EU Blue Card holder; Available from April 2026.

PROFESSIONAL EXPERIENCE

Strama-MPS GmbH, Straubing – Robotics Simulation Engineer

(October 2024 – March 2026)

- Robot motion planning, collision detection, offline programming and robot safe operation.
- Onsite robot commissioning with KRL to BMW manufacturing standards.
- Robot simulation for cycle time optimisation across 15+ projects.

Volkswagen AG, Wolfsburg – Master Thesis

(March.2023 – July.2023)

- Master Thesis on optimising industrial bin picking through Lean Simulation Methodology and PPR approach using Universal Robots, Python, RoboDK and CAD.

Otto-von-Guericke University, Magdeburg – Student Research Assistant

(December.2022 – March.2023)

- Research in association with Volkswagen AG on bin picking robot optimisation using KUKA, RoboDK and Python for robot simulation.

Otto-von-Guericke University, Magdeburg – Internship

(*October.2022 – March.2023)*

- Realisation of an Industry 4.0 based control system using PLC programming (IEC 61131) in WAGO PLC modules, OPC-UA concept, and Automation ML tool.

Envision Integrated Services Pvt. Ltd., India – Jr. Design Engineer

(August 2020 – January.2021)

- Engineering design projects using CATIA, Autodesk Inventor, Navisworks and AutoCAD.
- Digital factory conversion and CATIA Surfacing for automobiles.

SKILLS & KNOWLEDGE

- **Robotics & Automation:** Robot simulation (KUKA, ABB, FANUC, Staubli, UR), KUKA Certified Robot Programming (Prog I & II, Safe Operation), KUKA WorkVisual, Siemens Process Simulate, RoboGuide, RobotStudio, RoboDK, Staubli Robotics Suite, ROS2, Digital Twin, Virtual Commissioning, Industry 4.0.
- **CAD:** SolidWorks, CATIA V5, Autodesk Inventor, Creo, Solid Edge, AutoCAD.
- **Programming:** Python, KUKA-KRL, PLC IEC 61131, WAGO, e-Cockpit, Siemens S7 TIA.
- **Systems Engineering:** UML, SysML, AAS, Enterprise Architect, Automation ML, PLM, PTC Windchill, MBSE.

ACADEMIC QUALIFICATION

Otto-von-Guericke University, Magdeburg, Germany – Master of Science

(October.2021 – February.2024)

- Systems Engineering for Manufacturing
- Focus: Model Based Systems Engineering, Factory Automation & Industrial Robotics. (Grade: 2.1)

Visvesvaraya Technological University, Belagavi, India – Bachelor of Engineering

(July.2015 – July.2019)

- Mechanical Engineering
- Focus: Mechanical design, analysis, mechatronics & automation. (Grade: 2.1)

CERTIFICATIONS

- KUKA Robot Programming I (Basic)
- KUKA Robot Programming II (Advanced)
- KUKA Robot Safe Operation
- Universal Robot Programming – e Series Core Track

PROJECTS

- KUKA KRL Reader: KUKA program visualiser application using Python. Available on GitHub. (2026)
- Robot simulation for cycle time optimisation across projects involving Body-in-White, Automotive, E-Mobility, Medical Electronics; onsite commissioning of KUKA robots to BMW standards. (2024 – 2026)
- Master thesis: PPR-based digital factory planning for lean industrial robot operations. Backend simulation in RoboDK and Python for collision detection and reachability. Custom UI for 3D bin-picking data visualisation. (2023)
- Optimisation of industrial bin picking robot: gripper conceptualisation and robot motion planning. (2023)
- Development of collaborative robot based industrial operational concepts using Universal Robots. (2023)
- Realisation of an Industry 4.0 based control system using cyber-physical systems using model-based systems engineering and Asset Administration Shells. (2022–2023)
- Requirements engineering based design for a Zero Gravity Airplane concept using German VDI Standards. (2021–2022)
- Bachelor thesis: Generation of electricity and pyrolysis oil from plastic waste. (2018–2019)
- Design and development of a Hydro-Pneumatic Integrated Shaper Machine circuit. (2018)
- Technical paper presentation: Bionics – Prosthetic Technology. (2019)

ADDITIONAL EXPERIENCE

- Early Engineering Internships, India (2018–2020): Product design and GD&T (Gloinnt Solutions Pvt. Ltd); Industrial hydraulic and pneumatic systems (Rexroth Bosch Lab, DSU); Manufacturing processes and R&D in railway coach production (BEML).
- Certified training program on 3D Design and Printing Technology. (2017)
- Finalist, ISIE Indian Karting (IKR 2016–2017) Go-Kart Design & Development.

LANGUAGES & INTERESTS

- **Languages:** English (fluent), German (professional – B2), Hindi (fluent), Kannada (mother tongue).
- **Interests:** Travelling, Cooking, Hiking, Football, Reading, Animes and Documentaries.

Straubing, Germany

Thejas Dixit Sathyanarayana